

Alfredo Montes – Curriculum Vitae

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Accreditation and Training

The Flooring Education Symposium: Dalton, GA—October 2025

- Materials Testing
- The Basics of Moisture Testing
- IICRC Standards and Certifications
- Inspection, Assessment, and Evaluation
- The Art of Inspection – What Do You See?
- Causes and Remedies of Concrete Slab Imperfections

World of Concrete: Las Vegas, NV—January 2025

- ICRI Concrete Slab Moisture Testing Technician Certification – Grade 1:
 - ASTM F710 – 22
 - ASTM F1869 – 23
 - ASTM F2170 – 19a
 - ASTM F2659 – 23
 - ASTM F3191 – 23

Education

University of California, Irvine—September 2015 - June 2020

- B.S. in Materials Science and Engineering (MSE)

Relevant UCI Coursework

Statics

Addition and resolution of forces, distributed forces, equivalent system of forces, centroids, first moments, moments and products on inertia, equilibrium of rigid bodies, trusses, beams, and cables.

Mechanics of Structures

Introduction to material mechanics. Topics include stresses and strains, axial and torsional loading, bending, shear, combined loading, stress and strain transformation, failure theories, beam deflection, and buckling

<i>Communications in the Professional World</i>	Upper-division technical writing course including the development of presentation skills. Effective communication with a range of audiences. Recognition of ethical and professional responsibilities for engineers.
<i>Polymeric Materials</i>	An introduction to physical aspects of polymers, including polymer molecular size and configuration, solution and bulk phase structural properties, mechanical properties including viscoelasticity and rheology, polymer synthesis methods, characterization techniques, processing methods, and examples from existing and emerging polymeric materials.
<i>Mechanical Behavior and Design Principles</i>	An introduction to physical aspects of polymers, including polymer molecular size and configuration, solution and bulk phase structural properties, mechanical properties including viscoelasticity and rheology, polymer synthesis methods, characterization techniques, processing methods, and examples from existing and emerging polymeric materials.
<i>Materials Kinetics and Phase Transformations</i>	Treatment of the kinetics of solid-state reactions and reactions at interfaces. Thermodynamics and kinetics of phase transformations, including solidification processes, diffusional and diffusionless phase transformations.
<i>Ceramic Materials</i>	Microstructure and properties of ceramic materials.
<i>Design Failure Investigation</i>	Survey of mechanisms by which devices fail, including overload, fatigue, corrosion, and wear. Use of fractography and other evidence to interpret failure modes and specify design/manufacturing changes. Students redesign failed parts or structures based on actual parts and/or case histories.
<i>X-ray Diffraction, Electron Microscopy, and Microanalysis</i>	Material characterization using X-ray diffraction and scanning electron microscopy (SEM). Topics include X-ray diffraction and analysis; SEM imaging and microanalysis.
<i>Fundamentals of Materials Processing</i>	In-depth knowledge of manufacturing processes (from liquid, solid, and vapor phases) with focus on

the fundamentals of materials processing – transport and mechanical forces. Processing fundamentals that apply to the three main classes of engineering materials: metals, ceramics, and polymers.

Thermodynamics of Materials

Treatment of the laws of thermodynamics and their application in understanding properties and equilibrium states of engineering materials. Develops relationships pertaining to multiphase equilibrium and presents graphical constructions for the interpretation of phase diagrams. Statistical thermodynamics in relation to materials phenomena.

Materials Selection and Design

Meaning and phases of design; design considerations; properties of engineering materials; materials property charts; materials selection; process selection; multi-constraint and multi-objective design. Selection of shape in mechanical components. Designing with hybrid materials: challenges and opportunities. Environmental considerations: case studies.

Projects

American Football Helmet Pad using Lattices

UCI Senior Design Project—Fall 2019 - Spring 2020

Material Research for Lattice

- Utilized CES edupack software to find materials that fit within specific design constraints

Lattice Design

- Used Ntopology software to simulate force impacts 3d model of project football pad by inserting lattice model templates available in software

Work Experience

Blackbelt Collective Krav Maga - Fountain Valley, CA

Self Defense Instructor, July 2019 - Present

Elite Protective and Investigative Services Group - Various locations

Security Guard, February 2022 - November 2023

Paley Stone and Tile Forensics - Santa Ana, CA

Stone and Tile Flooring Inspector, February 2024 - Present